

IMAT MOCK EXAM 3 — "Genetics & Molecular Rotation"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 45/90.0 · Mock exam · answer key + worked solutions at the end

Section A - Logical Reasoning & Reading (Q1-9)

1. "A gene is a region of DNA that encodes a functional product, usually a protein. In eukaryotes, coding regions (exons) are interrupted by non-coding regions (introns). After transcription, introns are removed and exons are spliced together to form mature messenger RNA, which is then translated by ribosomes."

Which statement is not supported?

- A)** Exons are coding regions.
- B)** Ribosomes transcribe DNA into RNA.
- C)** Introns are removed before translation.
- D)** A gene typically encodes a protein.
- E)** Splicing produces mature mRNA.

2. "A mutation is a heritable change in the nucleotide sequence of a genome. Mutations may be neutral, harmful, or — rarely — beneficial. Beneficial mutations can increase in frequency in a population over generations through natural selection, provided they confer a reproductive advantage."

The passage implies:

- A)** All mutations are harmful.
- B)** Neutral mutations are eliminated quickly.
- C)** Mutation rate determines fitness directly.
- D)** Natural selection acts on existing variation, including beneficial mutations.
- E)** Beneficial mutations always become fixed.

3. (General knowledge) Which Italian scientist won the 1986 Nobel Prize in Physiology or Medicine for the discovery of nerve growth factor (NGF)?

- A)** Camillo Golgi
- B)** Carlo Rubbia
- C)** Rita Levi-Montalcini
- D)** Salvador Luria
- E)** Renato Dulbecco

4. In a race, Ali finished before Bashir but after Chen. Dev finished last. Who finished first?

- A)** Chen
- B)** Ali
- C)** Dev
- D)** Cannot be determined
- E)** Bashir

5. "All enzymes are proteins." This statement is:

- A)** True only for digestive enzymes
- B)** Always true
- C)** True only in prokaryotes
- D)** Always false
- E)** Mostly true, but some RNA molecules (ribozymes) also catalyse reactions

6. In a certain language, "TREE" is written as "USFF". How is "BIRD" written?

- A)** CJTE **B)** AJSE **C)** CJSF **D)** BKSE **E)** CJSE

7. What comes next: 3, 6, 11, 18, 27, ?

- A)** 36 **B)** 38 **C)** 39 **D)** 40 **E)** 41

8. If today is Tuesday, what day will it be 100 days from today?

- A)** Monday **B)** Wednesday **C)** Tuesday **D)** Thursday **E)** Friday

9. A clock reads 7:00. What is the angle between the hour and minute hands?

- A)** 0° **B)** 90° **C)** 150° **D)** 180° **E)** 210°

Section B - Biology (Q10-32)

10. In DNA, the two strands run in opposite directions. This property is called:

- A)** Antiparallelism
B) Methylation
C) Supercoiling
D) Complementarity
E) Semiconservative

11. DNA replication is described as "semiconservative" because:

- A)** Both strands of the parent molecule are preserved together.
B) Replication is conservative in prokaryotes only.
C) Only the leading strand is copied.
D) Each new molecule consists of one parental strand and one newly synthesised strand.
E) The entire molecule is built anew.

12. The enzyme that joins Okazaki fragments by forming phosphodiester bonds is:

- A)** Helicase
B) Topoisomerase
C) DNA ligase
D) Primase
E) DNA polymerase III

13. During transcription in eukaryotes, the enzyme responsible is:

- A)** DNA polymerase
B) RNA polymerase
C) Ligase
D) Reverse transcriptase
E) Helicase

14. During which process are introns removed from pre-mRNA?

- A)** Splicing (post-transcriptional modification)
B) DNA repair
C) Translation
D) Transcription (initiation)
E) Replication

15. The three-nucleotide sequence on mRNA that specifies an amino acid is a:

- A)** Gene **B)** Anticodon **C)** Operon **D)** Promoter **E)** Codon

16. The anticodon is found on:

- A)** SnRNA **B)** mRNA **C)** DNA **D)** tRNA **E)** rRNA

17. The bonds linking amino acids in a protein are:

- A)** Peptide bonds
B) Hydrogen bonds
C) Phosphodiester bonds
D) Ester bonds
E) Glycosidic bonds

- 18.** The secondary structure of proteins (α -helix, β -sheet) is stabilised mainly by:
- A)** Peptide bonds
 - B)** Hydrogen bonds between backbone atoms
 - C)** Ionic bonds
 - D)** Hydrophobic interactions
 - E)** Disulfide bridges
- 19.** A diploid cell with $2n = 6$ undergoes meiosis. How many chromosomes are in each resulting gamete?
- A)** 2 **B)** 3 **C)** 4 **D)** 6 **E)** 12
- 20.** In a cross between two heterozygous tall plants ($Tt \times Tt$), where T (tall) is dominant, what is the expected phenotypic ratio in the offspring?
- A)** 9:3:3:1
 - B)** All tall
 - C)** 3:1 (tall : short)
 - D)** 1:2:1
 - E)** 1:1
- 21.** In humans, haemophilia is an X-linked recessive trait. A carrier mother ($X^H X^h$) and an unaffected father ($X^H Y$) have a son. What is the probability the son has haemophilia?
- A)** 0 **B)** 25% **C)** 50% **D)** 75% **E)** 100%
- 22.** A man who is colour-blind (X-linked recessive) has daughters with a woman who is not colour-blind and is not a carrier. What is true of their daughters?
- A)** None will carry the allele.
 - B)** All will be colour-blind.
 - C)** All will be colour-blind and carriers.
 - D)** 50% colour-blind, 50% carriers.
 - E)** All will be carriers (heterozygous) but not colour-blind.
- 23.** In a dihybrid cross ($AaBb \times AaBb$) with independent assortment, the expected phenotypic ratio is:
- A)** 1:1 **B)** 1:2:1 **C)** All dominant **D)** 3:1 **E)** 9:3:3:1
- 24.** A bacterium acquires antibiotic resistance via a plasmid from another bacterium. This is an example of:
- A)** Meiosis
 - B)** Mutation
 - C)** Mitosis
 - D)** Conjugation (horizontal gene transfer)
 - E)** Transcription
- 25.** The phases of mitosis in correct order are:
- A)** Prophase, metaphase, anaphase, telophase
 - B)** Prophase, anaphase, metaphase, telophase
 - C)** Telophase, anaphase, metaphase, prophase
 - D)** Metaphase, prophase, anaphase, telophase
 - E)** Anaphase, metaphase, prophase, telophase
- 26.** Which organelle contains digestive enzymes and functions in intracellular digestion?
- A)** Mitochondrion **B)** Chloroplast **C)** Nucleolus **D)** Ribosome **E)** Lysosome

27. Which structure is common to both plant and animal cells?
A) Plasmodesmata
B) Chloroplast
C) Mitochondrion
D) Large central vacuole
E) Cell wall
28. In cellular respiration, glycolysis produces a net gain of:
A) 0 ATP (all invested)
B) 2 ATP, 2 NADH, 2 pyruvate (per glucose)
C) 2 ATP, 2 FADH₂, 2 pyruvate
D) 4 ATP, 4 NADH, 2 pyruvate
E) 38 ATP
29. The non-protein component required for some enzymes to function (e.g., a metal ion) is called a:
A) Cofactor (or coenzyme if organic)
B) Apoenzyme
C) Competitive inhibitor
D) Substrate
E) Allosteric inhibitor
30. At the neuromuscular junction, the neurotransmitter released by motor neurons is:
A) GABA **B)** Serotonin **C)** Glutamate **D)** Acetylcholine **E)** Dopamine
31. The resting potential of a typical neuron is approximately:
A) -70 mV **B)** -30 mV **C)** 0 mV **D)** +30 mV **E)** +70 mV
32. Which vitamin is required for blood clotting?
A) Vitamin D **B)** Vitamin A **C)** Vitamin K **D)** Vitamin C **E)** Vitamin B12

Section C - Chemistry (Q33-47)

33. How many moles are in 36 g of water (H₂O; molar mass = 18 g/mol)?
A) 0.5 **B)** 1 **C)** 2 **D)** 18 **E)** 36
34. Balance: $__ \text{C}_3\text{H}_8 + __ \text{O}_2 \rightarrow __ \text{CO}_2 + __ \text{H}_2\text{O}$. The coefficient of O₂ is:
A) 3 **B)** 4 **C)** 5 **D)** 6 **E)** 8
35. The oxidation number of nitrogen in NO₃⁻ is:
A) -3 **B)** +3 **C)** +4 **D)** +5 **E)** +6
36. What is the pH of a 0.001 M solution of HNO₃ (strong acid)?
A) 1 **B)** 2 **C)** 3 **D)** 11 **E)** 13
37. Which of the following is a weak acid?
A) HNO₃
B) HBr
C) HCl
D) H₂SO₄
E) CH₃COOH (acetic acid)
38. Which bond is most polar?
A) C-H **B)** O-H **C)** H-H **D)** C-C **E)** N-H

39. Boyle's law states that, at constant temperature, the pressure of a fixed mass of gas is:
A) Independent of volume
B) Inversely proportional to its volume
C) Proportional to temperature
D) Directly proportional to its volume
E) Proportional to moles
40. The number of protons in an atom of ^{23}Na (sodium) is:
A) 11 **B)** 12 **C)** 22 **D)** 23 **E)** 34
41. Which of the following is an alkene?
A) C_3H_8 **B)** C_6H_6 **C)** CH_4 **D)** C_2H_4 **E)** C_2H_6
42. The functional group of alcohols is:
A) $-\text{CHO}$ **B)** $-\text{NH}_2$ **C)** $-\text{COOH}$ **D)** $-\text{CO}-$ **E)** $-\text{OH}$
43. Which has a $\text{C}=\text{O}$ group bonded to two carbons (within a chain)?
A) Carboxylic acid
B) Ketone
C) Alcohol
D) Aldehyde
E) Amine
44. Increasing the concentration of a reactant in a reaction generally:
A) Decreases activation energy
B) Decreases the rate
C) Shifts K_{eq}
D) Increases the rate (more frequent collisions)
E) Has no effect on rate
45. For an endothermic reaction, ΔH is:
A) Positive
B) Zero
C) Equal to activation energy
D) Negative
E) Always equal to ΔS
46. Which isotope is radioactive?
A) ^{13}C **B)** ^{16}O **C)** ^{14}N **D)** ^{12}C **E)** ^{14}C
47. A solution has a pH of 8. It is:
A) Strongly basic
B) Strongly acidic
C) Weakly acidic
D) Neutral
E) Weakly basic

Section D - Mathematics (Q48-54)

48. Simplify: $(2^3) \times (2^2)$.
A) $2^5 (= 32)$ **B)** 2^6 **C)** $2^1 (= 2)$ **D)** 4^5 **E)** 8
49. Solve: $\log_{10}(x) = 2$.
A) $x = 20$ **B)** $x = 100$ **C)** $x = 10$ **D)** $x = 2$ **E)** $x = 1000$
50. Solve: $x^2 = 49$ ($x > 0$).
A) 6 **B)** 7 **C)** 8 **D)** 14 **E)** 24.5

51. Solve the inequality: $x/3 - 1 \geq 2$.
A) $x \geq 6$ B) $x \geq 1$ C) $x \geq 12$ D) $x \geq 9$ E) $x \geq 3$
52. A square has perimeter 20 cm. What is its area?
A) 5 cm² B) 10 cm² C) 20 cm² D) 25 cm² E) 100 cm²
53. $\cos 60^\circ = ?$
A) 0 B) $\sqrt{3}/2$ C) $1/2$ D) $\sqrt{2}/2$ E) 1
54. Two fair coins are tossed. What is the probability of getting at least one head?
A) $1/4$ B) $1/2$ C) 1 D) $1/3$ E) $3/4$

Section E - Physics (Q55-60)

55. A car decelerates uniformly from 20 m/s to rest in 5 s. What is its deceleration?
A) 2 m/s² B) 4 m/s² C) 5 m/s² D) 10 m/s² E) 20 m/s²
56. Three 6 Ω resistors are connected in parallel. The equivalent resistance is:
A) 2 Ω B) 3 Ω C) 6 Ω D) 9 Ω E) 18 Ω
57. What is the voltage across a 10 Ω resistor carrying 0.5 A?
A) 0.05 V B) 5 V C) 10 V D) 20 V E) 50 V
58. A ball of mass 0.5 kg moves at 4 m/s. What is its kinetic energy?
A) 1 J B) 2 J C) 4 J D) 8 J E) 16 J
59. A 10 N force moves an object 5 m in the direction of the force. How much work is done?
A) 0 J B) 2 J C) 5 J D) 15 J E) 50 J
60. Specific heat capacity of water is 4200 J/(kg·°C). How much energy heats 2 kg of water by 10 °C?
A) 8,400 J B) 42,000 J C) 84,000 J D) 420,000 J E) 840 J

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	B	2	D	3	C	4	A	5	E
6	E	7	B	8	D	9	C	10	A
11	D	12	C	13	B	14	A	15	E
16	D	17	A	18	B	19	B	20	C
21	C	22	E	23	E	24	D	25	A
26	E	27	C	28	B	29	A	30	D
31	A	32	C	33	C	34	C	35	D
36	C	37	E	38	B	39	B	40	A
41	D	42	E	43	B	44	D	45	A
46	E	47	E	48	A	49	B	50	B
51	D	52	D	53	C	54	E	55	B
56	A	57	B	58	C	59	E	60	C

Worked solutions

Each solution shows the reasoning that produces the key, and names the wrong options by their content. Letters refer to the option order printed in this paper.

Question 1 • Answer: B

The passage states that splicing produces mature mRNA which is then translated by ribosomes, so the only role given to ribosomes is translation. Transcription (DNA to RNA) is performed by RNA polymerase, not ribosomes, so the claim that ribosomes transcribe DNA into RNA is contradicted by the passage. Every other statement is directly supported: exons are called coding regions, introns are removed before translation, splicing forms mature mRNA, and a gene is said to usually encode a protein.

Question 2 • Answer: D

The passage says beneficial mutations can increase in frequency over generations through natural selection, which only makes sense if selection acts on variation that already exists in the population. The other claims all contradict or overstate the text: it explicitly allows neutral and beneficial mutations (so not all are harmful), says only that beneficial mutations can increase in frequency (not that they always become fixed), gives no reason for neutral mutations to be quickly eliminated, and links fitness advantage to the mutation's effect rather than to the mutation rate itself.

Question 3 • Answer: C

The 1986 Nobel Prize in Physiology or Medicine for the discovery of nerve growth factor (NGF) went to Rita Levi-Montalcini, shared with the American biochemist Stanley Cohen. The other Italians won in different fields and years: Golgi for his staining method revealing nerve cell structure (1906), Luria for phage genetics (1969), Dulbecco for tumour viruses (1975), and Rubbia for physics (1984), none connected to NGF.

Question 4 • Answer: A

Ali finished before Bashir but after Chen, which fixes the partial order Chen, then Ali, then Bashir. Dev is stated to be last, so only fourth place remains for him and the full order is Chen, Ali, Bashir, Dev. No other arrangement satisfies both statements, so Chen finished first.

Question 5 • Answer: E

Catalysis is not exclusive to proteins: ribozymes are RNA molecules that catalyse reactions, and the peptidyl transferase activity of the ribosome itself is RNA-based. So the blanket claim is mostly true but has known exceptions, exactly as stated in the answer. Calling it always true ignores ribozymes, while calling it always false ignores that the overwhelming majority of enzymes are proteins.

Question 6 • Answer: E

TREE becomes USFF and each letter is shifted forward by one position in the alphabet: T to U, R to S, and each E to F. Applying the same rule to BIRD gives B to C, I to J, R to S, D to E, spelling CJSE. No other candidate keeps every letter shifted by exactly one.

Question 7 • Answer: B

The gaps between consecutive terms are 3, 5, 7 and 9, the successive odd numbers, so the next gap is 11. The next term is $27 + 11 = 38$. Check: adding 3, 5, 7, 9 in turn to 3 gives 6, 11, 18, 27, confirming the pattern holds throughout.

Question 8 • Answer: D

Weekdays repeat with period 7, so only the remainder of 100 divided by 7 matters. Since $100 = 14 \times 7 + 2$, the remainder is 2, meaning 100 days from today falls on the same weekday as 2 days from today. Tuesday plus 2 days is Thursday.

Question 9 • Answer: C

The full dial is 360° divided into 12 hour marks, so each hour is $360 \div 12 = 30^\circ$. At 7:00 the minute hand points at 12 and the hour hand points at 7, i.e. $7 \times 30 = 210^\circ$ measured clockwise from 12. The angle between the hands is the smaller of 210° and $360 - 210 = 150^\circ$, so it is 150° .

Question 10 • Answer: A

The two strands of a DNA double helix run in opposite 5' to 3' directions: one strand runs 5' to 3' while its partner runs 3' to 5'. This geometric property is called antiparallelism. Complementarity instead refers to the A-T and G-C base pairing, supercoiling describes how the helix is coiled upon itself, and semiconservative describes the mode of replication.

Question 11 • Answer: D

In semiconservative replication each parental strand stays intact and acts as a template, so every daughter double helix contains one original strand and one newly synthesised strand. The Meselson and Stahl experiment confirmed exactly this. The conservative model, which would keep both parental strands together in one molecule and build a wholly new molecule in the other, is what the experiment ruled out.

Question 12 • Answer: C

Okazaki fragments are short pieces synthesised discontinuously on the lagging strand. DNA polymerase extends them but cannot close the final gap, so DNA ligase is the enzyme that seals the nick by forming the phosphodiester bond between adjacent fragments. Helicase unwinds the helix, primase lays RNA primers and topoisomerase relieves supercoiling, none of which joins fragments.

Question 13 • Answer: B

Transcription is the synthesis of RNA from a DNA template, catalysed by RNA polymerase. DNA polymerase copies DNA during replication, reverse transcriptase makes DNA from an RNA template in retroviruses, and helicase and ligase modify DNA structure rather than synthesising RNA.

Question 14 • Answer: A

The primary transcript (pre-mRNA) still contains introns, which are removed by the spliceosome during post-transcriptional processing called splicing; the exons are then joined into the mature mRNA. Translation reads only the finished message and never removes introns, and replication and DNA repair act on DNA, not on RNA.

Question 15 • Answer: E

The genetic code is read on mRNA in triplets, and each three-nucleotide sequence specifying one amino acid (or a stop signal) is a codon. The anticodon is the complementary triplet located on tRNA, while a promoter is the DNA region where transcription starts and an operon is a cluster of genes transcribed together.

Question 16 • Answer: D

tRNA carries the anticodon in its anticodon loop; this three-base sequence pairs with the complementary codon on mRNA so that the correct amino acid is delivered to the growing chain. mRNA carries the codons themselves and rRNA forms the structural and catalytic core of the ribosome.

Question 17 • Answer: A

The carboxyl group of one amino acid reacts with the amino group of the next, releasing water and forming a peptide bond (an amide linkage) between them. Phosphodiester bonds instead link nucleotides in nucleic acids, glycosidic bonds link sugars, and hydrogen bonds only stabilise the folded three-dimensional shape.

Question 18 • Answer: B

Secondary structure consists of regular arrangements of the polypeptide backbone, and both the alpha-helix and the beta-pleated sheet are stabilised by hydrogen bonds between backbone C=O and N-H groups. Disulfide bridges and hydrophobic interactions mainly stabilise the tertiary structure, and peptide bonds belong to the primary sequence.

Question 19 • Answer: B

$2n = 6$ means the diploid cell carries two sets of 3 chromosome types, six in total. Meiosis separates homologous pairs, halving the number, so each gamete receives one set, $n = 3$ chromosomes. Keeping 6 would describe mitosis, and 12 would be a doubling rather than a halving.

Question 20 • Answer: C

Step 1: the cross $Tt \times Tt$ produces genotypes TT, Tt, Tt, tt in a 1:2:1 ratio. Step 2: since T is dominant, TT and both Tt combinations are tall, so 3 of the 4 outcomes are tall and 1 is short, giving a phenotypic ratio of 3 tall : 1 short. The 1:2:1 figure is the genotypic ratio and does not describe phenotypes, while 9:3:3:1 belongs to a dihybrid cross.

Question 21 • Answer: C

A son receives the Y chromosome from his father, so his single X always comes from his mother. The carrier mother is $X^H X^h$, so each son has a 50% chance of receiving the X^h chromosome and being affected. The 25% figure would answer a different question, namely the chance per pregnancy of an affected child of unspecified sex.

Question 22 • Answer: E

The colour-blind father has only one X chromosome, and he passes that chromosome (bearing the recessive allele) to every daughter. The mother, who is neither colour-blind nor a carrier, always passes a normal X . Every daughter is therefore heterozygous $X^C X^N$: a carrier whose normal second X masks the recessive allele, so none of them is colour-blind.

Question 23 • Answer: E

Step 1: for one gene, $Aa \times Aa$ gives a 3/4 dominant to 1/4 recessive phenotypic split. Step 2: with independent assortment the two genes multiply: $(3/4)(3/4) = 9/16$ show both dominant traits, $(3/4)(1/4) = 3/16$ for each of the two single-recessive classes, and $(1/4)(1/4) = 1/16$ show both recessive traits. This yields the classic 9:3:3:1 ratio.

Question 24 • Answer: D

Receiving a plasmid from another bacterium is conjugation, the classic case of horizontal gene transfer: DNA moves sideways between two living cells rather than being inherited from a parent cell. Mutation is a change within a cell's own genome, while mitosis, meiosis and transcription are cell division or gene expression processes that do not transfer DNA between cells.

Question 25 • Answer: A

Mitosis proceeds prophase (chromosomes condense, spindle forms), metaphase (chromosomes align on the equatorial plate), anaphase (sister chromatids separate and move to the poles), telophase (nuclear envelopes reform). Any ordering that puts anaphase before metaphase, or telophase first, contradicts these events, so the sequence is prophase, metaphase, anaphase, telophase.

Question 26 • Answer: E

Lysosomes are membrane-bound vesicles filled with acid hydrolases that break down worn-out organelles and material taken into the cell, which is intracellular digestion. Ribosomes build proteins, mitochondria and chloroplasts produce ATP or sugars, and the nucleolus assembles ribosomal subunits.

Question 27 • Answer: C

Mitochondria, the site of aerobic respiration, occur in both plant and animal cells. The cell wall, the large central vacuole, chloroplasts and plasmodesmata are features of plant cells and are absent from animal cells, so only the mitochondrion is shared.

Question 28 • Answer: B

Glycolysis invests 2 ATP in its preparatory phase and regenerates 4 ATP in its payoff phase, for a net gain of 2 ATP, together with 2 NADH and 2 pyruvate molecules per glucose. The claim of 4 ATP quotes the gross production and forgets the 2 consumed, and $FADH_2$ is produced by the Krebs cycle, not by glycolysis.

Question 29 • Answer: A

Many enzymes need a non-protein helper to be active: an inorganic ion such as Zn^{2+} or Mg^{2+} is called a cofactor, while an organic helper molecule is a coenzyme. The apoenzyme is the inactive protein part without its helper, and the substrate is the molecule the enzyme acts on, so neither of those fits the description.

Question 30 • Answer: D

Motor neurons release acetylcholine at the neuromuscular junction; it binds receptors on the muscle fibre membrane and triggers depolarisation and contraction. Dopamine, GABA, serotonin and glutamate are neurotransmitters of the central nervous system, not the junction with skeletal muscle.

Question 31 • Answer: A

In a resting neuron the inside of the membrane sits about 70 mV below the outside, maintained by potassium leak channels and the Na^+/K^+ pump, so the resting potential is approximately -70 mV. The value $+30$ mV instead corresponds to the peak of an action potential, not to the resting state.

Question 32 • Answer: C

Vitamin K is required for the gamma-carboxylation of clotting proteins such as prothrombin (its very name comes from Koagulation), so without it blood fails to clot properly. Vitamin C is involved in collagen synthesis, vitamin D in calcium absorption, vitamin A in vision, and vitamin B12 in red blood cell formation.

Question 33 • Answer: C

Amount of substance follows $n = m \div M$. Substituting the given values: $n = 36 \text{ g} \div 18 \text{ g/mol} = 2 \text{ mol}$. The figure 18 is the molar mass itself, and 0.5 mol would correspond to only 9 g of water.

Question 34 • Answer: C

Step 1: balance carbon, 3 C in propane gives 3 CO_2 . Step 2: balance hydrogen, 8 H gives 4 H_2O . Step 3: balance oxygen, the products contain $3 \times 2 + 4 \times 1 = 10$ O atoms, so the reactant side needs $10 \div 2 = 5$ O_2 . The balanced equation $\text{C}_3\text{H}_8 + 5 \text{O}_2 \rightarrow 3 \text{CO}_2 + 4 \text{H}_2\text{O}$ checks on every element, so the oxygen coefficient is 5.

Question 35 • Answer: D

Each oxygen is assigned -2 , so three oxygens contribute $3 \times (-2) = -6$. If x is the oxidation number of nitrogen, the sum over the ion must equal its charge: $x + (-6) = -1$, giving $x = +5$.

Question 36 • Answer: C

HNO_3 is a strong acid, so it dissociates completely and $[\text{H}^+] = 0.001 \text{ M} = 10^{-3} \text{ M}$. pH is defined as $-\log_{10}[\text{H}^+] = -\log_{10}(10^{-3}) = 3$. The values 11 and 13 would describe strongly basic solutions (pH $14 - 3$), which is the mirror-image trap.

Question 37 • Answer: E

HCl , HNO_3 , H_2SO_4 and HBr are all strong acids that dissociate essentially completely in water. Acetic acid, CH_3COOH , is the only weak acid in the list: it establishes an equilibrium in which most molecules remain undissociated.

Question 38 • Answer: B

Bond polarity grows with the electronegativity difference between the two atoms. Using Pauling values, O (3.44) versus H (2.20) differs by about 1.2, N (3.04) versus H by about 0.8, C (2.55) versus H by about 0.4, while C-C and H-H join identical atoms and are completely non-polar. The O-H bond therefore shows the largest charge separation.

Question 39 • Answer: B

Boyle's law states $pV = \text{constant}$ for a fixed amount of gas at constant temperature, which means pressure is inversely proportional to volume: doubling the volume halves the pressure. Direct proportionality to volume contradicts the law, and proportionality to temperature or moles describes different, non-isothermal or non-fixed-amount conditions.

Question 40 • Answer: A

The mass number 23 counts protons plus neutrons and changes between isotopes, but the proton number is fixed by the element's atomic number. Sodium's atomic number is 11, so every sodium atom has 11 protons (and in this isotope $23 - 11 = 12$ neutrons). The answers 23 or 12 confuse the mass number or neutron count with the proton count.

Question 41 • Answer: D

Alkenes contain one C=C double bond and follow the general formula C_nH_{2n} . C_2H_4 (ethene) fits this formula exactly and is the alkene. CH_4 , C_2H_6 and C_3H_8 are saturated alkanes fitting C_nH_{2n+2} , and C_6H_6 is the aromatic benzene ring, which is not classified as an alkene.

Question 42 • Answer: E

Alcohols are defined by a hydroxyl group, -OH, bonded to a saturated carbon, as in ethanol CH_3CH_2OH . The -CHO group marks aldehydes, -COOH carboxylic acids, -CO- ketones, and -NH₂ amines.

Question 43 • Answer: B

A carbonyl group (C=O) whose carbon is bonded to two other carbons inside the chain is the defining feature of a ketone, as in propanone $CH_3-CO-CH_3$. In an aldehyde the carbonyl carbon is terminal and carries a hydrogen, and in a carboxylic acid it is additionally bonded to an -OH.

Question 44 • Answer: D

A higher reactant concentration means more particles per unit volume, so collisions occur more frequently and more particles per second possess the energy and orientation needed to react, raising the rate. Concentration does not change the activation energy, which is a fixed property of the reaction pathway, and it does not shift the equilibrium constant K, which only temperature can change.

Question 45 • Answer: A

ΔH measures the enthalpy change of the system; endothermic means the system absorbs heat, so the products end up higher in enthalpy than the reactants and ΔH is positive. The sign convention is the reverse for exothermic reactions. ΔH also cannot equal ΔS (a quantity in J/K) or the activation energy, which measures the barrier height rather than the overall energy change.

Question 46 • Answer: E

Carbon-14 contains two more neutrons than the stable isotopes ^{12}C and ^{13}C and decays by beta emission with a half-life of about 5730 years, which is the basis of radiocarbon dating. ^{16}O and ^{14}N are stable nuclides, so carbon-14 is the only radioactive species listed.

Question 47 • Answer: E

At 25 °C neutrality corresponds to pH 7, where $[H^+] = 10^{-7}$ M. A pH of 8 means $[H^+] = 10^{-8}$ M, ten times fewer hydrogen ions than neutral water, so the solution is mildly (weakly) basic. Strongly basic solutions lie well above pH 10, and anything below 7 would be acidic.

Question 48 • Answer: A

When multiplying powers with the same base, exponents add: $2^3 \times 2^2 = 2^{3+2} = 2^5 = 32$. Arithmetic check: $8 \times 4 = 32$, confirming the result. The value 2^6 adds one exponent too many, 4^5 wrongly changes the base instead of adding exponents, and 8 is just 2^3 alone.

Question 49 • Answer: B

$\log_{10}(x) = 2$ asks for the number whose base-10 logarithm is 2, i.e. 10 raised to the power 2. Converting from log to exponential form gives $x = 10^2 = 100$. Check: $\log_{10}(100) = \log_{10}(10 \times 10) = 2$.

Question 50 • Answer: B

$x^2 = 49$ has two real solutions, $x = 7$ and $x = -7$, since both square to 49. The condition $x > 0$ discards the negative root, leaving $x = \sqrt{49} = 7$. The value 14 mistakes doubling for square-rooting, and 24.5 divides 49 by 2 instead of taking the root.

Question 51 • Answer: D

Step 1: add 1 to both sides, giving $x/3 \geq 3$. Step 2: multiply both sides by 3; since 3 is positive the inequality direction is unchanged, so $x \geq 9$. Boundary check: at $x = 9$ the left side is $9/3 - 1 = 2$, which satisfies the $\geq$ condition, and any larger x works while any smaller x fails.

Question 52 • Answer: D

A square has four equal sides, so the side length is $20 \div 4 = 5$ cm. The area is side squared: $5 \times 5 = 25$ cm². The value 100 cm² comes from wrongly squaring the perimeter, and 20 cm² confuses the perimeter with the area.

Question 53 • Answer: C

In a right triangle with angles 30°, 60°, 90° the sides opposite them are in the ratio 1 : $\sqrt{3}$: 2. Cosine is adjacent over hypotenuse, and for 60° the adjacent side is the shortest leg, so $\cos 60^\circ = 1/2$. The value $\sqrt{3}/2$ belongs to $\cos 30^\circ$, the classic mix-up.

Question 54 • Answer: E

Two coin tosses give four equally likely outcomes: HH, HT, TH, TT. Exactly one outcome (two tails) contains no head, so $P(\text{no head}) = 1/4$ and $P(\text{at least one head}) = 1 - 1/4 = 3/4$. Alternatively, HT, TH and HH are the three favourable outcomes out of four. The value 1/2 treats the event as if only one toss mattered.

Question 55 • Answer: B

Step 1: use $a = \Delta v \div t = (v_{\text{final}} - v_{\text{initial}}) \div t$. Step 2: substitute the data: $(0 - 20) \div 5 = -4$ m/s². The negative sign indicates slowing down, so the deceleration has magnitude 4 m/s².

Question 56 • Answer: A

For resistors in parallel the reciprocals add: $1/R = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$, so $R = 2 \Omega$. Equivalent shortcut: n equal resistors R in parallel give R/n , here $6/3 = 2 \Omega$. The value 18Ω is the series combination, the classic trap for this arrangement.

Question 57 • Answer: B

Ohm's law gives $V = I \times R$. Substituting: $V = 0.5 \text{ A} \times 10 \Omega = 5 \text{ V}$. The value 0.05 V results from dividing the current by the resistance instead of multiplying, and 50 V moves the decimal point the wrong way.

Question 58 • Answer: C

Kinetic energy is $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.5 \text{ kg} \times (4 \text{ m/s})^2 = 0.25 \times 16 = 4 \text{ J}$. The value 2 J comes from computing mv without squaring the speed, and 16 J would be mv^2 without the factor of $\frac{1}{2}$.

Question 59 • Answer: E

Work equals force times the distance moved in the direction of the force: $W = F \times d = 10 \text{ N} \times 5 \text{ m} = 50 \text{ J}$. The value 2 J comes from dividing instead of multiplying, and 0 J would only be correct if the force acted perpendicular to the motion.

Question 60 • Answer: C

Use the heating relation $Q = m \times c \times \Delta T$ with $m = 2 \text{ kg}$, $c = 4200 \text{ J/(kg} \cdot ^\circ\text{C)}$ and $\Delta T = 10 ^\circ\text{C}$: $Q = 2 \times 4200 \times 10 = 84,000 \text{ J}$ (84 kJ). The value 42,000 J drops the factor of 2 from the mass, and 840 J is a decimal-place error.